

Birthdays, Schooling, and Crime: New Evidence on the Dropout-crime Nexus

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Motivation

- Theoretical Evidence: education on crime
 - Becker 1968; Ehrlich 1973
 - relative payoff to legitimate activities
 - peer effect
- Empirical Evidence: compulsory schooling on criminal activity.
 - Lochner and Moretti 2004; Anderson 2009; Hjalmarsson, Holmlund and Lindquist 2011.etc

Research Question & Main Results

- **Research Question:** How does school entry eligibility affect children's subsequent educational and criminal outcomes?
- **Main Results:**
 - Children born after the cut date:
 - perform better in reading and math in middle school
 - less likely to be involved in juvenile delinquency
 - more likely to drop out of high school and commit felony by age 19
 - Higher dropout rate is due to longer exposure to legal possibility of dropping out

Rest of Our Talk

- 1 Introduction
- 2 Data Description
- 3 Research Design
- 4 Estimation Results
- 5 Robustness Check

Literature Review

- Positive effect of relative maturity on education outcome?
 - Bedard and Dhuey, 2006; Datar, 2006.etc
 - Popularity of “Academic Redshirting ”
 - Disappear in later years?(Elder and Lubotsky 2009)
- Negative effect of late entry to school on life outcomes?
 - Angrist and Krueger, 1991; Dobkin and Ferreira, 2010; McCray and Royer, 2011
- This research: school entry eligibility → variation in education attainment → crime

Institution Background

- Minimum age requirement for school entry in North Carolina
 - Eligible to enter kindergarten: 5 years old by Oct.16th
 - Children born on or after Oct.17th have to wait another year.
 - Example: Oct.16, 1990 → 1995
 - Oct.17, 1990 → 1996
- Minimum school leaving age in North Carolina: 16
 - Exposure to legal possibility of dropping out
 - Children born on Oct.16th: 19 months
 - Children born on Oct.17th: 31 months
 - Example: Oct.16,1990 → May, 2008
 - Oct.17,1990 → May, 2009

Real-life Complication

- Distribution of students across grades is more complicated
 - Private kindergartens can set their own minimum age policies
 - Students may attend kindergarten from a different state
 - Kindergarten attendance is not required
- Other qualifications:
 - Parents delay their child's enrollment in school by a year: "Academic Redshirting"
 - Students are retained in primary school by school officials
- Intent-to-treat(ITT) effects: effect of receiving school entry eligibility one year earlier on educational and criminal outcomes

Data Source & Outcome Variables

- North Carolina birth certificate data
 - merged with educational data via NCERDC individual identifiers
 - contains exact date of birth, key demographic attributes
 - 120-day window
- Individual-level administrative data from North Carolina public school system, NCERDC
 - EOG reading and math tests from grade 3 to 8
 - Enrollment data between grade 3 and 12
 - Potential complication: composition change? → restrict sample
- Official juvenile complaint, NCDJJDP and Adult felony conviction data, NCDOC

Descriptive Statistics

- 1987-1988, 1991-1993 cohorts
- 120-day, 60-day window

Table 1: Descriptive Statistics

Cohort:	1987-1988		1991-1993	
Samples born within X Days of Cutoff Date:	120 Days	60 Days	120 Days	60 Days
Male	49.0%	49.3%	48.6%	48.7%
Black	30.6%	30.3%	32.0%	31.9%
Unwed mother	26.0%	25.8%	33.2%	33.0%
Born to Teenage Mother	10.3%	10.1%	10.5%	10.6%
Mother's Education < High School	12.5%	12.5%	11.7%	11.7%
Birth Weight < 2500 grams	7.3%	7.2%	7.7%	7.7%
Free/Reduced Price Lunch , Age 11			43.8%	43.6%
Grade Level, Age 11	5.3	5.3	5.3	5.3
Grade Level, Age 15	9.3	9.2	9.3	9.2
Grade 6 Reading Percentile	49.2	49.4	50.0	50.3
Grade 7 Reading Percentile	48.4	48.6	49.8	50.0
Grade 8 Reading Percentile	48.5	48.7	49.4	49.6
Grade 6 Math Percentile	50.1	50.3	50.8	51.0
Grade 7 Math Percentile	49.7	50.0	50.7	50.8
Grade 8 Math Percentile	49.5	49.6	51.4	51.5
Grade Retention, Age 11-15	8.4%	8.1%	5.6%	5.4%
School Enrollment, 3 Years after Grade 8	89.4%	89.3%		
School Enrollment, 4 Years after Grade 8	79.0%	78.8%		
Juvenile Delinquency Age 13-15			9.0%	8.8%
Criminal Conviction, Age 17-19	5.7%	5.8%		
Obs.	69572	35213	110244	55318

Note: Sample is composed of students born around the school entry cutoff dates in 1987-1988 and 1991-1993. Each cohort is defined by students born between June 19 of the given year and February 14 of the following year (120 days before and after the cutoff date of October 17 (= Day 0), respectively). The sample is further restricted to students 1) who attended NC public schools between age 11 and 15, 2) whose NC birth certificate information is available.

- **Identification Strategy:** Regression Discontinuity(Fuzzy RD)
- **Identification Assumption:** Other characteristics associated with educational and criminal outcomes are comparable and continuous at the cutoff date

Regression Model

- Global parameter specification:

$$Y_i = \alpha_1 D_i + \alpha_2 R_i + \alpha_3 D_i R_i + \alpha_4 R_i^2 + \alpha_5 D_i R_i + X_i \beta + \epsilon_i$$

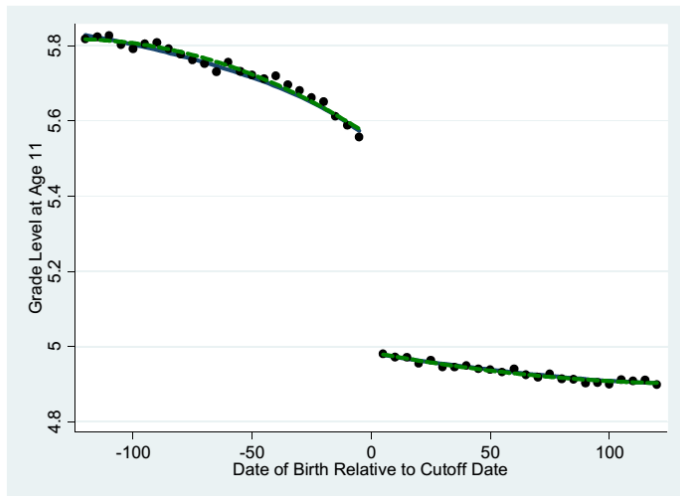
- Local linear regressions:

$$\min \sum_{i=1}^{N_i} (Y_i - \alpha_1 D_i - \alpha_2 R_i - \alpha_3 D_i R_i - X_i \beta)^2 K_h(D_i, R_i)$$

where $K_h(D_i, R_i)$ is the weighting kernel function (triangular kernel) with bandwidth h .

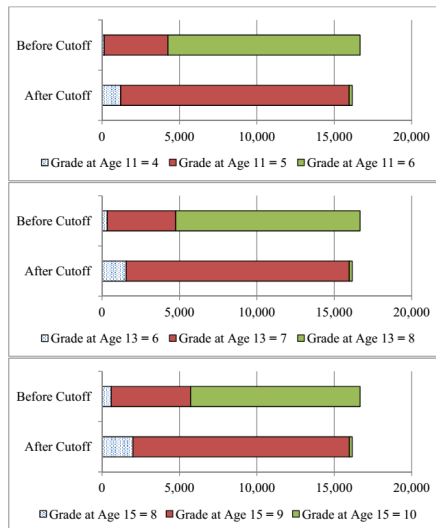
Grade Level Distribution at Age 11

Figure 2: Grade Level Distribution at Age 11, 1987-1988 Cohort



Grade Level Distribution

Figure 1: Grade Level Distribution across Age Levels, 1987 Cohort



Comment: Drop Figure 1

Grade Level Discontinuity at the Cutoff Date

Table 2: Effect of School Entry Eligibility on Grade Level Configuration

Outcome: Grade Level at	Age 11		Age 15	
(A) 1987-1988 Cohort				
D	-0.578*** (0.013)	-0.579*** (0.012)	-0.520*** (0.013)	-0.521*** (0.012)
R (x100)	-0.403*** (0.042)	-0.397*** (0.039)	-0.374*** (0.041)	-0.369*** (0.037)
DR (x100)	0.303*** (0.045)	0.299*** (0.042)	0.289*** (0.045)	0.295*** (0.042)
Male		-0.122*** (0.011)		-0.171*** (0.013)
Black		0.039*** (0.009)		-0.009 (0.009)
Unwed Mother		-0.062*** (0.006)		-0.103*** (0.008)
Teenage Mother		-0.051*** (0.009)		-0.093*** (0.011)
Mother's Ed. < HS		-0.108*** (0.008)		-0.195*** (0.011)
Low Birthweight		-0.076*** (0.011)		-0.089*** (0.013)
Constant	5.559*** (0.012)	5.651*** (0.013)	9.450*** (0.012)	9.607*** (0.015)
R-squared	0.408	0.430	0.291	0.339
Obs.	34,560	34,560	34,560	34,560

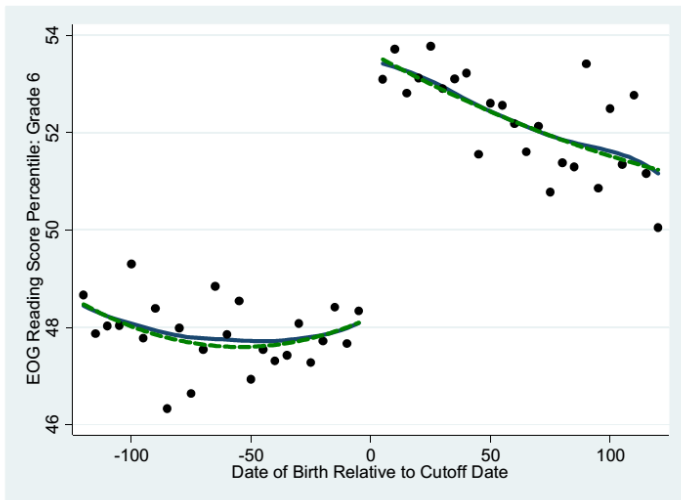
- Showing results with and without controls and see coefficients stay relatively unchanged is good;
- However, since R-square does not increase a lot, we worry if these controls are meaningful or not, and what will happen if they include meaningful controls.
- And it also weakens our confidence in the robustness check.

Grade Level Discontinuity at the Cutoff Date

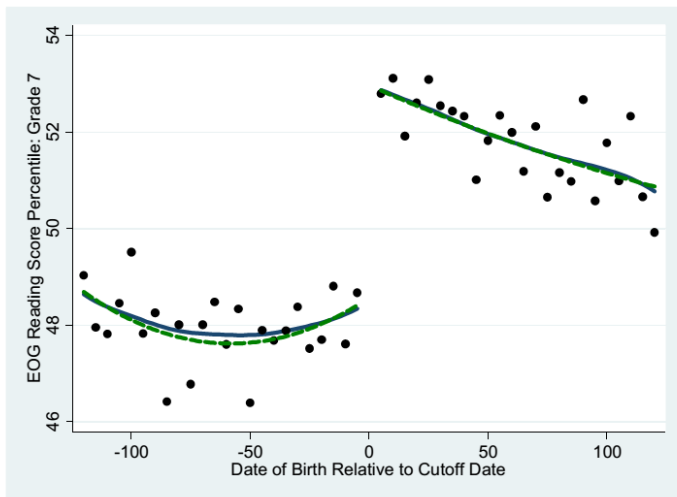
(B) 1991-1993 Cohort				
D	-0.567*** (0.013)	-0.567*** (0.012)	-0.511*** (0.013)	-0.510*** (0.011)
R (x100)	-0.327*** (0.038)	-0.322*** (0.034)	-0.309*** (0.038)	-0.304*** (0.033)
DR (x100)	0.221*** (0.042)	0.219*** (0.039)	0.197*** (0.043)	0.194*** (0.038)
Male		-0.124*** (0.008)		-0.158*** (0.010)
Black		0.052*** (0.007)		0.036*** (0.007)
Unwed Mother		-0.047*** (0.007)		-0.068*** (0.007)
Teenage Mother		-0.054*** (0.009)		-0.072*** (0.010)
Mother's Ed. < HS		-0.133*** (0.008)		-0.171*** (0.009)
Low Birthweight		-0.084*** (0.011)		-0.084*** (0.011)
Free/Reduced Lunch		-0.113*** (0.006)		-0.148*** (0.006)
Constant	5.522*** (0.011)	5.652*** (0.011)	9.444*** (0.011)	9.619*** (0.013)
R-squared	0.342	0.379	0.274	0.337
Obs.	54,416	54,416	54,416	54,416

Discontinuity in Students' Achievement Levels

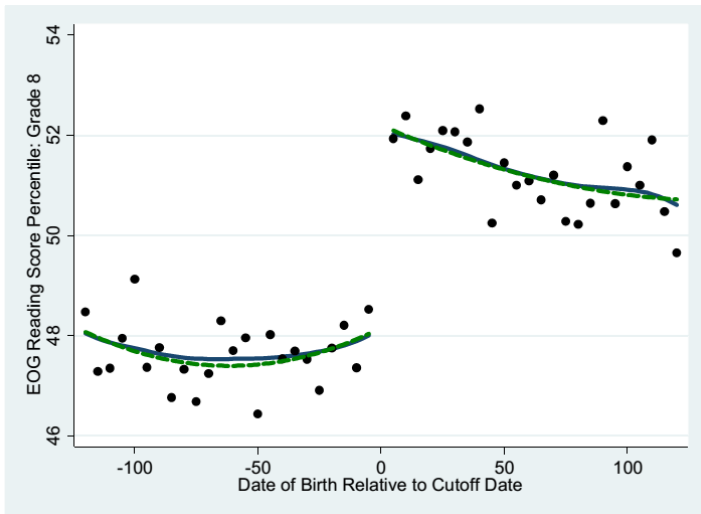
Figure 3: EOG Reading Score Percentiles in Grades 6 through 8, 1987-1988 Cohort



Discontinuity in Students' Achievement Levels

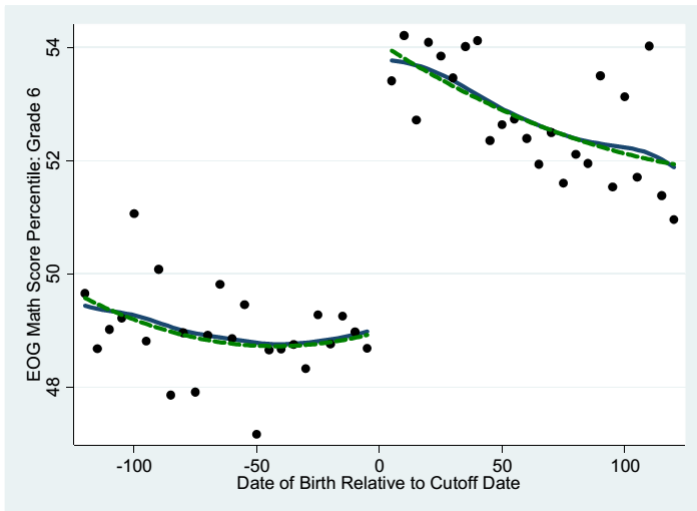


Discontinuity in Students' Achievement Levels

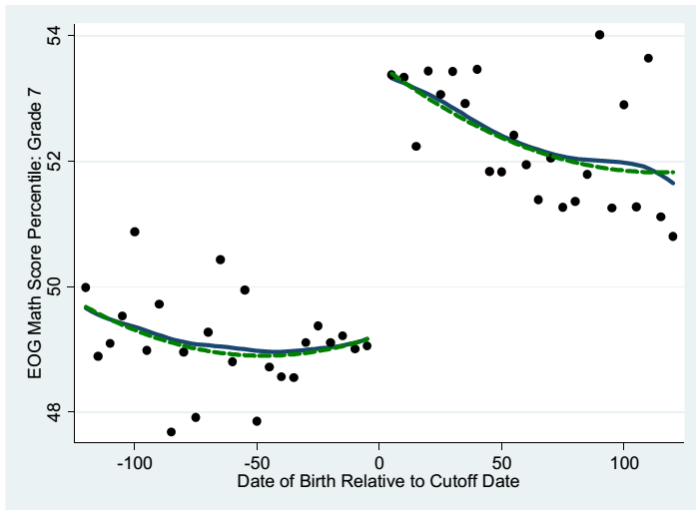


Discontinuity in Students' Achievement Levels

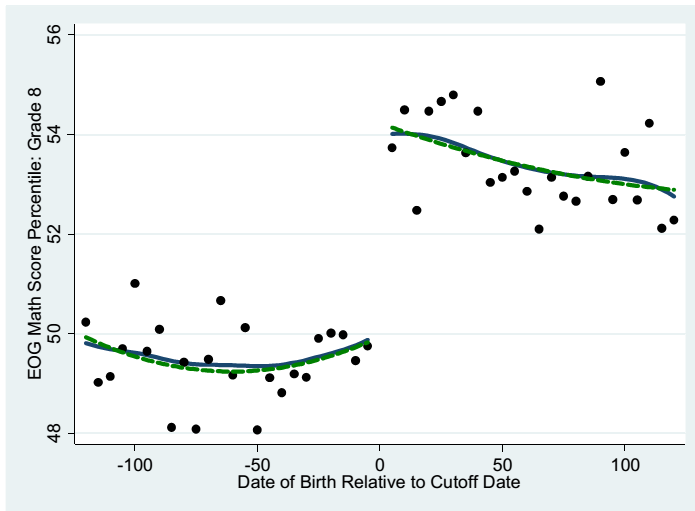
Figure 4: EOG Math Score Percentiles in Grades 6 through 8, 1987-1988 Cohort



Discontinuity in Students' Achievement Levels



Discontinuity in Students' Achievement Levels



Discontinuity in Reading Percentile

Table 3: Effect of School Entry Eligibility on EOG Reading Achievement

Cohort:	1987-1988			1991-1993		
Outcome:	EOG Reading Percentile			EOG Reading Percentile		
	Grade 6	Grade 7	Grade 8	Grade 6	Grade 7	Grade 8
D	6.250*** (0.615)	6.001*** (0.592)	4.708*** (0.512)	5.487*** (0.450)	4.700*** (0.503)	4.178*** (0.580)
R	-0.021 (0.014)	-0.010 (0.014)	-0.005 (0.012)	0.017* (0.010)	0.017 (0.011)	0.016 (0.010)
DR	0.017 (0.020)	-0.007 (0.019)	0.001 (0.017)	-0.032** (0.014)	-0.033** (0.016)	-0.024 (0.018)
Male	-4.504*** (0.323)	-4.066*** (0.285)	-3.903*** (0.292)	-4.837*** (0.237)	-4.760*** (0.257)	-3.653*** (0.231)
Black	-17.244*** (0.418)	-17.779*** (0.393)	-18.310*** (0.433)	-13.507*** (0.347)	-12.582*** (0.339)	-13.047*** (0.314)
Unwed Mother	-7.924*** (0.475)	-7.503*** (0.430)	-7.485*** (0.501)	-4.969*** (0.325)	-5.440*** (0.316)	-5.599*** (0.317)
Teenage Mother	-7.824*** (0.537)	-8.103*** (0.489)	-8.082*** (0.536)	-5.010*** (0.430)	-5.043*** (0.435)	-5.050*** (0.420)
Mother's Ed. < HS	-15.952*** (0.504)	-16.116*** (0.529)	-15.858*** (0.497)	-10.540*** (0.345)	-11.181*** (0.345)	-10.580*** (0.327)
Low Birthweight	-2.786*** (0.681)	-2.183*** (0.707)	-2.525*** (0.686)	-1.366*** (0.495)	-1.058** (0.409)	-0.633 (0.414)
Free/Reduced Lunch				-12.653*** (0.208)	-12.313*** (0.261)	-12.845*** (0.268)
Constant	58.981*** (0.471)	58.461*** (0.498)	58.723*** (0.431)	63.071*** (0.444)	63.022*** (0.420)	62.257*** (0.442)
R-squared	0.198	0.199	0.204	0.245	0.233	0.240
Obs.	34,560	34,560	34,560	54,416	54,416	54,416

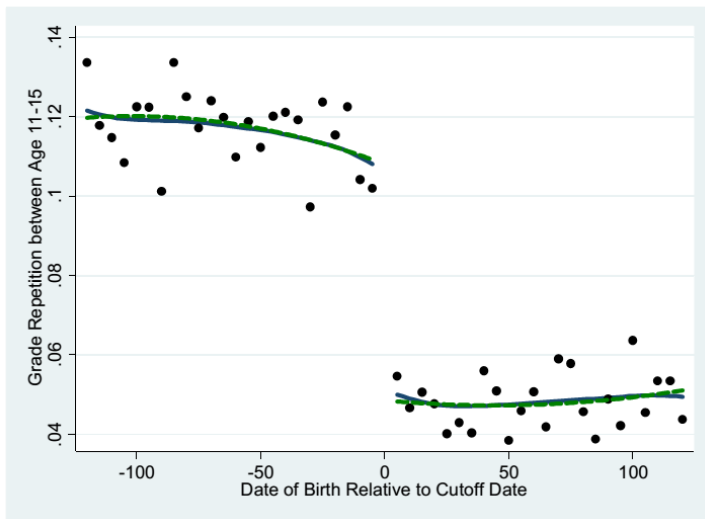
Discontinuity in Reading Percentile

Table 3: Effect of School Entry Eligibility on EOG Reading Achievement

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Outcome:	EOG Reading Percentile			EOG Reading Percentile		
	Grade 6	Grade 7	Grade 8	Grade 6	Grade 7	Grade 8
D	6.250*** (0.615)	6.001*** (0.592)	4.708*** (0.512)	5.487*** (0.450)	4.700*** (0.503)	4.178*** (0.580)
R	-0.021 (0.014)	-0.010 (0.014)	-0.005 (0.012)	0.017* (0.010)	0.017 (0.011)	0.016 (0.010)
DR	0.017 (0.020)	-0.007 (0.019)	0.001 (0.017)	-0.032** (0.014)	-0.033** (0.016)	-0.024 (0.018)
Male	-4.504*** (0.323)	-4.066*** (0.285)	-3.903*** (0.292)	-4.837*** (0.237)	-4.760*** (0.257)	-3.653*** (0.231)
Black	-17.244*** (0.418)	-17.779*** (0.393)	-18.310*** (0.433)	-13.507*** (0.347)	-12.582*** (0.339)	-13.047*** (0.314)
Unwed Mother	-7.924*** (0.475)	-7.503*** (0.430)	-7.485*** (0.501)	-4.969*** (0.325)	-5.440*** (0.316)	-5.599*** (0.317)
Teenage Mother	-7.824*** (0.537)	-8.103*** (0.489)	-8.082*** (0.536)	-5.010*** (0.430)	-5.043*** (0.435)	-5.050*** (0.420)
Mother's Ed. < HS	-15.952*** (0.504)	-16.116*** (0.529)	-15.858*** (0.497)	-10.540*** (0.345)	-11.181*** (0.345)	-10.580*** (0.327)
Low Birthweight	-2.786*** (0.681)	-2.183*** (0.707)	-2.525*** (0.686)	-1.366*** (0.495)	-1.058** (0.409)	-0.633 (0.414)
Free/Reduced Lunch				-12.653*** (0.208)	-12.313*** (0.261)	-12.845*** (0.268)
Constant	58.981*** (0.471)	58.461*** (0.498)	58.723*** (0.431)	63.071*** (0.444)	63.022*** (0.420)	62.257*** (0.442)
R-squared	0.198	0.199	0.204	0.245	0.233	0.240
Obs.	34,560	34,560	34,560	54,416	54,416	54,416

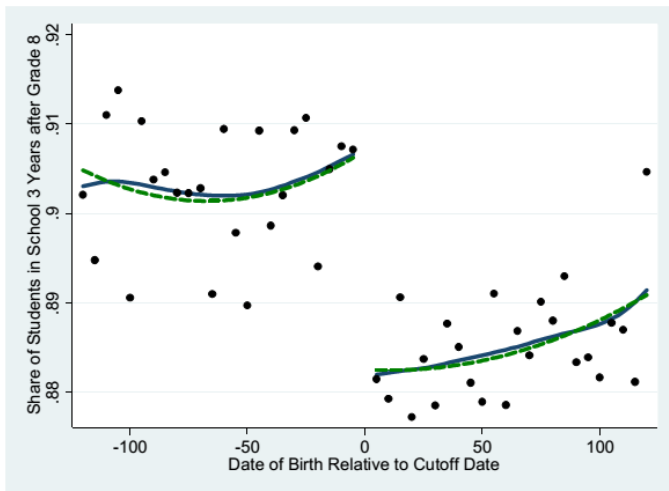
Discontinuity in Likelihood of Repeating Grade

Figure 5: Share of Students Repeating Grades between Age 11-15, 1987-1988 Cohort

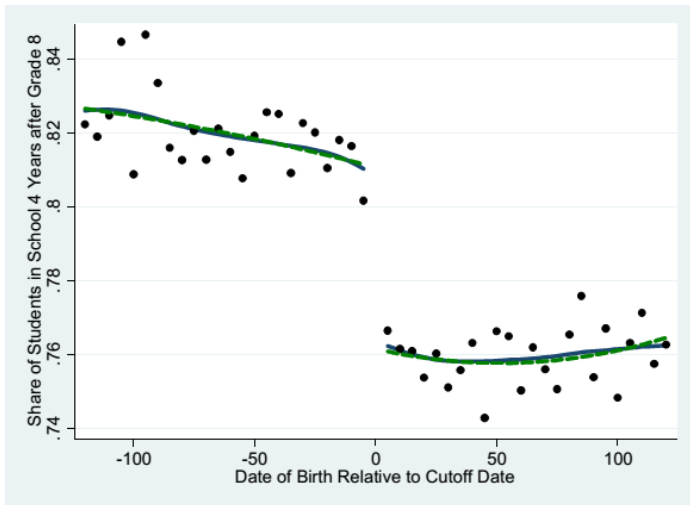


Discontinuity in Enrollment in Year 11

Figure 6: Share of Students Enrolled in School in Years 11 and 12, 1987-1988 (



Discontinuity in Enrollment in Year 12



Discontinuity in Enrollment?

Table 4: Effect of School Entry Eligibility on Grade Retention and High School Attainment

Cohort:	1987-1988		1991-1993	
Outcome:	Grade Retention	Enrollment	Enrollment	Grade Retention
	Age 11-15	Year 11	Year 12	Age 11-15
D	-5.712*** (0.482)	-2.386*** (0.781)	-4.448*** (0.773)	-5.588*** (0.400)
R	-0.026* (0.014)	0.011 (0.019)	-0.030* (0.018)	-0.018 (0.012)
DR	0.004 (0.017)	-0.003 (0.026)	0.014 (0.026)	0.022* (0.013)
Male	4.800*** (0.388)	-2.336*** (0.367)	-3.328*** (0.494)	3.519*** (0.296)
Black	4.753*** (0.495)	4.062*** (0.547)	7.056*** (0.706)	1.609*** (0.317)
Unwed Mother	4.134*** (0.571)	-5.572*** (0.639)	-8.127*** (0.795)	2.084*** (0.367)
Teenage Mother	4.181*** (0.630)	-8.098*** (0.986)	-11.145*** (1.082)	1.916*** (0.482)
Mother's Ed. < HS	8.477*** (0.735)	-14.366*** (0.761)	-17.471*** (0.792)	4.080*** (0.460)
Low Birthweight	1.217** (0.613)	-1.322* (0.733)	0.211 (0.935)	0.013 (0.445)
Free/Reduced Lunch				3.578*** (0.325)
Constant	4.332*** (0.546)	94.435*** (0.643)	88.078*** (0.793)	3.270*** (0.477)
R-squared	0.053	0.039	0.045	0.048
Obs.	34,560	34,560	34,560	54,416

Discontinuity in Juvenile Delinquency

Table 5: Effect of School Entry Eligibility on Juvenile Delinquency

Cohort:	1991-1993			
Outcome:	Age 13-15	Age 13	Age 14	Age 15
D	-2.803*** (0.571)	-0.443 (0.316)	-1.474*** (0.394)	-1.774*** (0.394)
R	0.028* (0.015)	0.006 (0.009)	0.020** (0.010)	0.016 (0.010)
DR	-0.001 (0.019)	-0.001 (0.010)	-0.006 (0.013)	0.006 (0.013)
Male	4.581*** (0.304)	1.454*** (0.152)	2.297*** (0.163)	2.650*** (0.226)
Black	1.002*** (0.353)	0.422* (0.229)	0.194 (0.229)	0.907*** (0.273)
Unwed Mother	3.994*** (0.453)	1.399*** (0.239)	1.916*** (0.301)	2.485*** (0.314)
Teenage Mother	3.686*** (0.732)	0.855** (0.349)	1.376*** (0.421)	2.393*** (0.519)
Mother's Ed. < HS	6.217*** (0.592)	2.167*** (0.345)	3.407*** (0.412)	2.810*** (0.428)
Low Birthweight	-1.191** (0.568)	-0.189 (0.355)	-0.354 (0.369)	-1.062*** (0.390)
Free/Reduced Lunch	6.641*** (0.379)	1.807*** (0.171)	3.073*** (0.234)	3.671*** (0.292)
Constant	2.855*** (0.488)	0.245 (0.266)	1.052*** (0.307)	1.577*** (0.342)
R-squared	0.049	0.016	0.024	0.028
Obs.	54,416	54,416	54,416	54,416

Discontinuity in Adult-felony-conviction

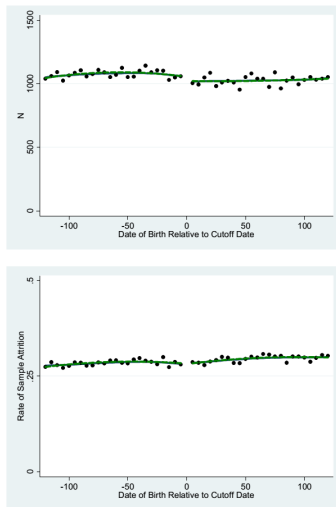
Table 6: Effect of School Entry Eligibility on Adult Criminal Conviction

Cohort:	1987-1988			
Outcome: Crime Committed at	Age 17-19	Age 17	Age 18	Age 19
D	0.851* (0.469)	-0.033 (0.288)	-0.174 (0.313)	0.849** (0.339)
R	-0.016 (0.010)	-0.000 (0.006)	0.001 (0.009)	-0.017** (0.008)
DR	0.013 (0.015)	0.006 (0.010)	0.010 (0.011)	0.010 (0.012)
Male	6.634*** (0.291)	2.566*** (0.177)	3.394*** (0.210)	3.172*** (0.192)
Black	2.596*** (0.397)	0.973*** (0.257)	1.334*** (0.270)	1.036*** (0.270)
Unwed Mother	4.303*** (0.442)	1.789*** (0.292)	2.014*** (0.287)	1.875*** (0.292)
Teenage Mother	2.906*** (0.561)	1.309*** (0.406)	1.512*** (0.423)	1.134*** (0.408)
Mother's Ed. < HS	3.955*** (0.497)	1.281*** (0.273)	2.308*** (0.392)	1.454*** (0.289)
Low Birthweight	-0.508 (0.586)	-0.217 (0.318)	-0.500 (0.353)	-0.266 (0.406)
Free/Reduced Lunch	-0.865** (0.393)	-0.312 (0.214)	-0.584** (0.290)	-0.526* (0.281)
Constant	0.039	0.017	0.022	0.016
Obs.	34,560	34,560	34,560	34,560

- Showing figures for main results, and also showing estimates with and without controls will be nice.

Testing identification assumption

Figure 8: Birth Frequency and Sample Attrition Rate



Testing Identification Assumption

Table 10: Continuity of Covariates at the Cutoff

[illegible]

Insensitivity to Bandwidth Selection?

Table 11: Robustness to Bandwidth Choices

Bandwidth =	10	30	60
(A) 1987-1988 Cohorts			
Grade 6 Reading Percentile	6.128*** (1.500)	5.971*** (0.772)	6.250*** (0.615)
Grade 7 Reading Percentile	7.524*** (1.609)	5.915*** (0.846)	6.001*** (0.592)
Grade 8 Reading Percentile	4.806*** (1.412)	4.438*** (0.668)	4.708*** (0.512)
Grade Retention, Age 11-15	-4.066*** (0.626)	-4.575*** (0.567)	-5.712*** (0.482)
School Enrollment, Year 12	-1.824* (0.984)	-3.521*** (0.961)	-4.448*** (0.773)
Crime, Age 17-19	1.221 (0.787)	0.853 (0.526)	0.851* (0.469)
Crime, Age 17	-0.218 (0.513)	-0.420 (0.344)	-0.067 (0.286)
Crime, Age 18	0.482 (0.823)	-0.089 (0.418)	-0.201 (0.312)
Crime, Age 19	0.622 (0.593)	1.032** (0.437)	0.817** (0.341)
Obs.	5,329	17,051	34,560

Insensitivity to Bandwidth Selection?

(B) 1991-1993 Cohorts			
Grade 6 Reading Percentile	3.732*** (0.751)	4.797*** (0.559)	5.487*** (0.450)
Grade 7 Reading Percentile	3.234*** (0.730)	4.187*** (0.637)	4.700*** (0.503)
Grade 8 Reading Percentile	2.520*** (0.813)	3.522*** (0.745)	4.178*** (0.580)
Grade Retention, Age 11-15	-4.118*** (0.467)	-4.977*** (0.437)	-5.588*** (0.400)
Juvenile Offense, Age 13-15	0.052 (1.014)	-2.321*** (0.845)	-2.803*** (0.571)
Juvenile Offense, Age 13	-0.124 (0.494)	-0.534 (0.425)	-0.443 (0.316)
Juvenile Offense, Age 14	-0.292 (0.813)	-1.342** (0.593)	-1.474*** (0.394)
Juvenile Offense, Age 15	0.435 (0.538)	-1.434** (0.548)	-1.774*** (0.394)
Obs.	8,579	26,922	54,416

- From what we see from the table, the estimates of school retention, school enrollment, and most importantly, juvenile offense at age 14 and 15, are all sensitive to bandwidth choice.

Exposure to Legal Dropout on Crime?

Table 12: Exposure to Legal Dropout, High School Attainment and Crime

Cohort:	1987-1988			1991-1993		
Sample:	"On-time"		Full	"On-time"		Full
Enrollment, Year 12	-10.377*** (0.771)	-10.136*** (0.755)	-4.448*** (0.773)			
Crime, Age 17-19	1.046** (0.506)	1.035** (0.501)	0.851* (0.469)			
Delinquency, Age 13-15				-2.366*** (0.687)	-2.509*** (0.674)	-2.803*** (0.571)
Reading Score, Age 11	No	Yes	No	No	Yes	No
Obs.	27,486	27,486	34,560	41,188	41,188	54,418

- The validity of non-randomly select the "on-time" group is challenged.
- Can not separate compliers and always-takers in the treatment group.
- Can not separate compliers and never-takers in the control group.
- Control group will not be a good counterfactual for treatment group anymore.

Thank you!